

OBSERVATORIYA SANGLOK, Tajikistan

WMO: 388440

Lat: 38.2614N

Lon: 69.2175E

Elev: 2239

StdP: 77.16

Time Zone: 5.00 (E05)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.4	-11.6	-22.5	0.7	-8.8	-21.0	0.8	-7.8	7.8	-4.2	7.1	-4.0	3.0	50	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	25.7	14.2	24.7	13.6	23.6	12.9	15.2	23.6	14.4	23.0	13.7	22.6	2.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
11.7	11.3	21.3	10.4	10.4	20.1	9.4	9.7	19.0	51.4	23.2	48.4	23.5	46.2	22.6	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.6	5.7	5.0	-15.5	29.2	2.9	2.6	-17.5	31.1	-19.2	32.6	-20.8	34.1	-22.9	36.0
			-16.0	17.4	2.9	1.9	-18.0	18.7	-19.7	19.8	-21.3	20.9	-23.4	22.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-4.0	-3.8	2.1	7.2	12.2	17.7	20.7	19.1	15.3	8.8	1.2	-1.7	(6)	
	DBStd	9.30	2.85	4.60	3.55	3.39	2.81	3.04	2.01	1.89	2.66	3.79	4.29	3.55	(7)	
	HDD10.0	1867	436	386	246	93	11	0	0	2	67	264	363		(8)	
	HDD18.3	3928	694	619	504	334	189	46	4	13	95	295	513	622	(9)	
	CDD10.0	1126	0	0	1	9	79	231	333	282	159	31	1	0	(10)	
	CDD18.3	144	0	0	0	0	0	26	79	37	3	0	0	0	(11)	
	CDH23.3	275	0	0	0	0	1	33	188	52	1	0	0	0	(12)	
	CDH26.7	13	0	0	0	0	0	3	8	2	0	0	0	0	(13)	
(14) Wind	WSAvg	2.3	2.8	2.7	2.6	2.5	2.3	2.0	1.9	1.8	1.8	2.1	2.5	2.7	(14)	
(15) Precipitation	PrecAvg	519	54	71	84	80	60	25	14	7	7	23	42	52	(15)	
	PrecMax	736	97	121	135	187	120	54	37	21	19	50	112	100	(16)	
	PrecMin	326	22	23	24	26	11	9	5	3	1	4	12	7	(17)	
	PrecStd	101	21	23	27	31	29	11	7	4	4	11	25	25	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	8.7	12.6	17.2	20.7	26.5	27.9	25.9	22.6	19.2	13.6	9.8	(19)	
		MCWB	-1.9	1.4	4.8	8.5	11.7	16.1	15.5	14.0	12.3	7.9	6.3	1.1	(20)	
	2%	DB	2.7	5.8	10.2	14.8	19.1	24.0	26.4	24.7	21.2	17.3	10.4	6.6	(21)	
		MCWB	-2.3	-0.1	3.6	7.4	11.2	13.1	14.8	13.6	11.3	7.5	3.4	0.0	(22)	
	5%	DB	1.2	3.6	8.6	13.5	17.9	22.9	25.2	23.6	20.2	15.6	8.3	4.8	(23)	
		MCWB	-3.4	-1.2	2.6	6.6	10.4	12.7	13.9	12.9	10.7	6.6	2.0	-1.0	(24)	
	10%	DB	-0.5	1.8	7.1	12.0	16.9	21.9	24.4	22.6	19.3	14.1	6.4	3.0	(25)	
		MCWB	-4.2	-2.1	1.8	5.9	9.5	12.3	13.4	12.5	10.1	6.0	0.5	-1.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.7	2.6	6.1	10.1	13.1	16.2	16.9	15.5	13.8	9.5	6.8	2.4	(27)	
		MCDB	1.6	7.0	9.0	14.1	19.2	24.7	26.4	24.3	21.5	16.6	12.6	7.3	(28)	
	2%	WB	-1.8	0.8	4.6	8.6	11.8	14.9	15.8	14.3	12.5	8.1	4.9	0.9	(29)	
		MCDB	1.0	5.4	8.1	12.5	17.6	21.9	25.2	23.2	20.1	15.6	9.4	5.8	(30)	
	5%	WB	-2.6	-0.7	3.5	7.7	10.9	13.8	14.6	13.8	11.5	7.1	3.0	-0.5	(31)	
		MCDB	0.1	2.2	7.4	11.1	16.7	21.4	23.6	22.7	19.1	14.4	6.5	3.2	(32)	
	10%	WB	-3.4	-1.6	2.6	7.0	10.1	12.9	14.0	13.1	10.7	6.4	1.6	-1.5	(33)	
		MCDB	-1.0	1.1	6.1	10.4	15.7	20.6	23.5	21.8	18.3	13.2	4.7	2.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	5.8	5.2	5.5	6.1	6.2	6.2	6.2	6.5	6.1	4.9	4.7	(35)	
		MCDBR	6.8	7.6	6.7	7.6	7.2	6.5	7.2	7.1	7.0	7.6	7.0	6.5	(36)	
	5% WB	MCWBR	4.8	5.3	5.1	5.4	5.1	4.5	5.0	4.6	4.4	4.3	4.6	4.1	(37)	
		MCWBR	5.9	6.3	5.9	6.5	6.6	6.8	7.1	6.6	6.8	7.3	6.7	5.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.288	0.322	0.364	0.370	0.378	0.399	0.395	0.370	0.342	0.358	0.330	0.294	(40)	
		taud	2.363	2.258	2.170	2.206	2.214	2.136	2.140	2.217	2.264	2.149	2.213	2.313	(41)	
	Edn at Noon		893	898	890	906	901	879	879	895	902	839	820	851	(42)	
			88	112	136	140	142	153	152	137	122	124	102	87	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.11	2.75	3.92	5.26	6.51	7.64	7.75	7.15	5.95	4.05	2.45	1.88	(44)	
	RadStd		0.24	0.42	0.35	0.56	0.49	0.34	0.24	0.24	0.19	0.37	0.33	0.23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	-0.36	-0.68	N/A	N/A	N/A	N/A

Nomenclature: See separate page